

$$\sqrt{1.4} \quad \int_{\Gamma} \frac{ds}{y-x}, \quad \Gamma - \text{orp. } (0; 2), (4, 0)$$

$$y = \frac{1}{2}x - 2, \quad 0 \leq x \leq 4, \quad x = 2y + 4, \quad -2 \leq y \leq 0$$

$$\int_{\Gamma} \frac{dx}{y-x} = -\sqrt{5} \int_{-2}^0 \frac{dy}{y+4} = \underline{-\sqrt{5} \ln 2}$$

$$\sqrt{9} \quad \int_{\Gamma} (x^{4/3} + y^{4/3}) ds, \quad x^{2/3} + y^{2/3} = a^{2/3}$$

$$x = a \cos^3 t$$

$$y = a \sin^3 t$$

$$0 < t < 2\pi$$

$$\begin{aligned} \int_{\Gamma} &= 4 \int_{\Gamma'} (x^{4/3} + y^{4/3}) ds = 4 \cdot \frac{3}{2} a \cdot \frac{1}{2} a^{4/3} \int_0^{\pi/2} (1 + \cos^2 2t) \sin 2t dt = \\ &= \left\{ \begin{array}{l} u = \cos 2t \\ du = -2 \sin 2t dt \end{array} \right\} = \int_0^1 (1 + u^2) du = \underline{4a^{7/3}} \end{aligned}$$

$$\sqrt{28.4} \quad \int_{\Gamma} (x^2 + y^2) dx + (x^2 - y^2) dy$$

$$\int_{\Gamma} = I_1 + I_2 + I_3$$

$$AB: x=0, 0 \leq y \leq 1 \rightarrow I_1 = \int_0^1 y^2 dy = \frac{1}{3}$$

$$BC: y=0, 0 \leq x \leq 1 \rightarrow I_2 = \int_0^1 x^2 dx = \frac{1}{3}$$

$$CA: y=1-x, 0 \leq x \leq 1 \rightarrow I_3 = -\int_0^1 (x^2 + (1-x)^2 - x^2 + (1-x)^2) dx = \underline{-\frac{1}{3}}$$

$$\text{от } \frac{1}{3}$$

N/85,4 $\Gamma: y^2=x, A(1,1), B(4,2), p(x,y)=y$
 $m = \int_{\Gamma} p(x,y) ds = \int_1^2 y \sqrt{1+(2y)^2} dy = \frac{1}{2} \int_1^2 \sqrt{1+4y^2} dy^2 =$
 $= \frac{1}{12} (17\sqrt{17} - 5\sqrt{5})$

N/110 $F=(4x-5y; 2x+y)$

1) AP: $x=1, -9 \leq y \leq -3$

$A_1 = \int_{-9}^{-3} (2x+y) dy = -24$

$A = A_1 + A_2 = 22$

$A_2 = \int_1^3 (4x+5) dx = 46$ (PB: $y=-3, 1 \leq x \leq 3$)

2) AQ: $y=-9, 1 \leq x \leq 3$ QB: $x=3, -9 \leq y \leq 3$

$A_1 = \int_1^3 (4x+5) dx = 106$

$A_2 = \int_{-9}^3 (6+y) dy = 0$

$A = A_1 + A_2 = 106$

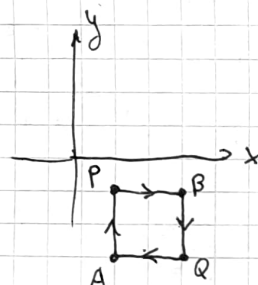
N/46 $\int (2x+y-y) dx$

$Q=x^2 \frac{\partial Q}{\partial x} = 2x$

$P=2xy-y \frac{\partial P}{\partial y} = 2x-1$

$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$

$\int_{\Gamma} \int_S dx dy = \pi ab$



N/100

Рассм $I = \int_{\partial G^+} (Ax+By) dx + (Cx+Dy) dy$

$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = C-B \rightarrow$

$\rightarrow I = (C-B) \iint_G dx dy \rightarrow S = \frac{I}{C-B}$ при $A=B=D=0, C=1: S = \int_{\partial G^+} x dy$
при $A=C=D=0, B=1: S = \int_{\partial G^+} y dx$

Т.к. $\int_{\partial G^+} x dy = - \int_{\partial G^+} y dx$ $S = \frac{1}{2} \int_{\partial G^+} x dy - y dx$

N/59 $\int_{\Gamma} 2xy dx + x^2 dy = \int_{\Gamma} d(x^2 y), u = x^2 y \rightarrow u(A)=0 \rightarrow u(B)=-4$

$\rightarrow \int_{\Gamma} = u(B) - u(A) = -4$

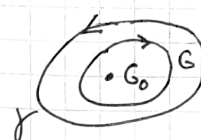
Т.к.

Т.к. $\Gamma_{уст} \partial G = \gamma$

1) $(0,0) P = -\frac{y}{x^2+y^2} \rightarrow \frac{\partial P}{\partial y} = \frac{y^2-x^2}{(x^2+y^2)^2}$

$Q = \frac{x}{x^2+y^2} \rightarrow \frac{\partial Q}{\partial x} = \frac{y^2-x^2}{(x^2+y^2)^2} \rightarrow \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} = 0 \rightarrow \int_{\gamma} = 0$

2)



G_0 - центр обн. $(0,0)$

$G_1 = G \setminus G_0$

$\int_{\gamma} + \int_{\partial G_0} = \int_{\gamma} = 0 \rightarrow \int_{\gamma} = \int_{\partial G_0} \rightarrow x = r \cos t \rightarrow y = r \sin t$
 $\rightarrow \int_{\gamma} = \int_0^{2\pi} \frac{r^2 \cos^2 t + r^2 \sin^2 t}{r^2} dt = \int_0^{2\pi} dt = 2\pi$

N37 $G = \{(x-1)^2 + y^2 \leq 1\} - \{x=y=z\} \subset R=1$

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2+y^2}}; \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2+y^2}}$$

$$S = \iint_G \sqrt{1 + \frac{x^2+y^2}{x^2+y^2}} dx dy = \sqrt{2} \iint_G dx dy = \frac{\pi \sqrt{2}}{1.6}$$

N51

$$[\bar{r}_\rho, \bar{r}_\psi] = \begin{vmatrix} \bar{e}_1 & \bar{e}_2 & \bar{e}_3 \\ -(b+a \cos \psi) \sin \rho & (b+a \cos \psi) \cos \rho & 0 \\ -a \sin \psi \cos \rho & -a \sin \psi \sin \rho & a \cos \psi \end{vmatrix} =$$

$$= \bar{e}_1 a(b+a \cos \psi) \cos \rho \cos \psi + \bar{e}_2 a(b+a \cos \psi) \sin \rho \cos \psi + \bar{e}_3 a(b+a \cos \psi) \sin \psi$$

$$|[\bar{r}_\rho, \bar{r}_\psi]|^2 = a^2(b+a \cos \psi)^2$$

$$S = \int_0^{2\pi} d\rho \int_0^{2\pi} a(b+a \cos \psi) d\psi = 2\pi a b 2\pi = 4\pi^2 a b$$

T2

$$\begin{aligned} x &= R \cos \rho \cos \psi & R \sin \psi_1 &= c \rightarrow 0 \leq \rho \leq 2\pi \\ y &= R \sin \rho \cos \psi & R \sin \psi_2 &= c+h \rightarrow \psi_1 \leq \psi \leq \psi_2 \\ z &= R \sin \psi \end{aligned}$$

$$S = \int_0^{2\pi} d\rho \int_{\psi_1}^{\psi_2} R^2 \cos \psi d\psi = 2\pi R^2 \left(\frac{c+h}{R} - \frac{c}{R} \right) = 2\pi R h$$

N1.1

$$x = 4 - 2y - 4z \rightarrow \iint_S (x+y+z) dS = \frac{1}{\sqrt{21}} \iint_G (4-y-3z) dy dz =$$

$$\frac{\partial x}{\partial y} = -2 \quad \frac{\partial x}{\partial z} = -4$$

$$= \frac{7}{3} \sqrt{21}$$

N18

$$\begin{aligned} x &= R \cos \rho \cos \psi \\ y &= R \sin \rho \cos \psi \\ z &= R \sin \psi \end{aligned} \quad G = \{0 \leq \rho \leq \frac{\pi}{2}, 0 \leq \psi \leq \frac{\pi}{2}\}$$

$$M = \iint_S dS = \iint_G R^2 \cos \psi d\rho d\psi = \frac{\pi R^2}{2}$$

$$x_c = \frac{1}{M} \iint_S x dS = \frac{R}{2}$$

$$x_c = y_c = z_c = \frac{R}{2}$$

N37

$$\begin{aligned} x &= a \cos \rho \cos \psi \\ y &= b \sin \rho \cos \psi \\ z &= c \sin \psi \end{aligned}$$

$$= a b c^2 \sin^2 \rho \cos^3 \psi$$

$$\int \int y z d\psi dx = \frac{1}{4} \pi a b c^2$$

$$\begin{vmatrix} 0 & b \sin \rho \cos \rho \cos \psi & 0 \\ -a \sin \rho \cos \psi & b \cos \rho \cos \psi & 0 \\ -a \cos \rho \sin \psi & -b \sin \rho \sin \psi & c \cos \psi \end{vmatrix} =$$

N38

$$x^2 = y^2 + z^2, \text{ гукл. замена } \begin{matrix} x = z \\ y = x \end{matrix} \rightarrow z^2 = x^2 + y^2$$

$$x = r \cos p$$

$$y = r \sin p$$

$$\iint_S = - \iint_G (2(x^2 + y^2) + x^2 + y^2) dx dy = -\frac{3}{2} \pi H^4$$

N42

$$\begin{matrix} x = r \cos p \\ y = r \sin p \end{matrix}, \begin{matrix} 0 \leq p \leq 2\pi \\ 0 \leq r \leq 1 \end{matrix}$$

$$\iint_S = - \iint_G \begin{vmatrix} x^6 & y^4 & (x^2 + y^2)^2 \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix} dx dy = -\frac{\pi}{3}$$

N44.1

$$\text{grad } f = (12x^3 + y, 3y^2 + x)^T$$

$$\text{grad } f(M) = (14, 13)^T$$

$$\vec{I} = \frac{\overline{AM}}{AM} = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$$

$$\frac{\partial f}{\partial \vec{e}} = (\text{grad } f(M), \vec{I}) = -\frac{14}{\sqrt{2}} + \frac{13}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$$

N48

$$\text{grad } f = (y^2 - 12x^3y^5, 2xy - 15x^4y^4)^T$$

$$\text{grad } f(M) = (-11, -13)^T$$

$$\vec{I} = (\cos \alpha, \sin \alpha)^T \rightarrow \frac{\partial f}{\partial \vec{I}} = (\vec{I}, \text{grad } f) \leq |\text{grad } f|$$

$$= \sqrt{290} \text{ - Max}$$

N13

$$\text{grad } f(u) = \left(\frac{\partial f(u)}{\partial x}, \frac{\partial f(u)}{\partial y}, \frac{\partial f(u)}{\partial z}\right)^T = \left(f'(u) \frac{\partial u}{\partial x}, f'(u) \frac{\partial u}{\partial y}, f'(u) \frac{\partial u}{\partial z}\right)^T = f'(u) \text{grad } u$$

N19

$$\nabla f(r) = \text{grad } f(r) = f'(r) \text{grad } r = \frac{f'(r)}{\sqrt{x^2 + y^2 + z^2}} (x\vec{i} + y\vec{j} + z\vec{k}) = f'(r) \frac{\vec{r}}{r}$$

N15

$$3) \text{grad } u = -\frac{1}{r^2} \frac{\vec{r}}{r} = -\frac{\vec{r}}{r^3}$$

$$5) u = (a_x \cdot x, a_y \cdot y, a_z \cdot z)^T \rightarrow \text{grad } u = \vec{a}$$

$$6) u = [\vec{a}, \vec{b}, \vec{r}] \rightarrow \text{grad } u = [\vec{a}, \vec{b}]$$

N37

$$\text{div}(u\vec{a}) = (\nabla, u\vec{a}) = (\nabla u, \vec{a}) + (\nabla \vec{a}, u) = (\text{grad } u, \vec{a}) + u \text{div} \vec{a}$$

$$\text{div}(u\vec{a}) = \frac{\partial(ua_x)}{\partial x} + \frac{\partial(ua_y)}{\partial y} + \frac{\partial(ua_z)}{\partial z} = a_x \frac{\partial u}{\partial x} + a_y \frac{\partial u}{\partial y} + a_z \frac{\partial u}{\partial z} + u \left(\frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z} \right) = (\text{grad } u, \vec{a}) + u \text{div} \vec{a}$$

N39

$$\text{div grad } u = (\nabla, \text{grad } u) = (\nabla, \nabla u) = (\nabla, \nabla) u = \Delta u =$$

$$= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

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$$\operatorname{div}(u \operatorname{grad} u) = (\nabla, u \nabla u) = (\nabla_u u, \nabla u) + u (\nabla_u, \nabla u) = \underbrace{(\nabla u)^2}_+ + u \Delta u$$

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$$3) \operatorname{div} \bar{r} = (\nabla, \bar{r}) = (\nabla_r, \bar{r}) = \frac{1}{r} (\bar{r}, \bar{r})$$

$$6) \operatorname{div} (f(\bar{r}), \bar{c}) = (\nabla_r, f(r), \bar{c}) = f'(r) \cdot (\bar{r}, \bar{c})$$

$$7) \operatorname{div} [\bar{c}, \bar{r}] = \frac{\partial (c_y z - c_z y)}{\partial x} + \frac{\partial (c_z x - c_x z)}{\partial y} + \frac{\partial (c_x y - c_y x)}{\partial z} = 0$$

143

$$3) \operatorname{rot}(u \bar{a}) = [\nabla, u \bar{a}] = [\nabla_u u, \bar{a}] + u [\nabla_{\bar{a}}, \bar{a}] = [\operatorname{grad} u, \bar{a}] + u \operatorname{rot} \bar{a}$$

$$5) \operatorname{rot} [\bar{a}, \bar{b}] = [\nabla, [\bar{a}, \bar{b}]] = \bar{a} \cdot (\nabla, \bar{b}) - \bar{b} \cdot (\nabla, \bar{a}) = \bar{a} \operatorname{div} \bar{b} - \bar{b} \operatorname{div} \bar{a}$$

$$6) \operatorname{div} [\bar{a}, \bar{b}] = (\nabla, [\bar{a}, \bar{b}]) = (\nabla_{\bar{a}}, \bar{a}, \bar{b}) + (\nabla_{\bar{b}}, \bar{a}, \bar{b}) = (\bar{b}, \operatorname{rot} \bar{a}) + (\bar{a}, \operatorname{rot} \bar{b})$$

150

$$\operatorname{rot}(u(r), \bar{r}) = [\nabla, u \bar{r}] = [\nabla_u u, \bar{r}] + u [\nabla_{\bar{r}}, \bar{r}] = \frac{u}{r} [\bar{r}, \bar{r}] + u [\nabla_{\bar{r}}, \bar{r}] = \underline{\underline{0}}$$

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$$\operatorname{rot} [\bar{r}, [\bar{c}, \bar{r}]] = \operatorname{rot} (\bar{c} r^2 - \bar{r} (\bar{r}, \bar{c})) = 2r [\frac{\bar{r}}{r}, \bar{c}] - [\bar{c}, \bar{r}] = 3 [\bar{r}, \bar{c}]$$

145

$$2) \bar{a} = (5x+y, 0, z)^T, \operatorname{div} \bar{a} = 5+0+1=6$$

$$x = 2r \cos \varphi \cos \psi$$

$$y = 3r \sin \varphi \cos \psi \rightarrow y = 6r^2 \cos \psi$$

$$z = r \sin \psi$$

$$\iiint_{\Omega} = -6 \iiint_{\Omega} dx dy dz = -6 \cdot 6 \int_0^{2\pi} d\varphi \int_{-\pi/2}^{\pi/2} \cos \psi d\psi \int_0^1 r^2 dr = -48\pi$$

$$3) x = r \cos \varphi \cos \psi \quad y = r^2 \cos \psi$$

$$y = r \sin \varphi \cos \psi$$

$$z = r \sin \psi$$

$$\iiint_{\Omega} = 6 \int_0^{2\pi} d\varphi \int_{-\pi/2}^{\pi/2} \cos \psi d\psi \int_1^2 r^2 dr = 56\pi$$

$$\sqrt{52} \quad \left\{ \begin{array}{l} x^2 + y^2 = z^2, \quad 0 \leq z \leq 4 \end{array} \right.$$

$$\vec{a} = (x^2, y^2, z^2)^T, \quad \text{div } \vec{a} = 2(x+y+z)$$

$$\left\{ \begin{array}{l} x = r \cos \varphi \\ y = r \sin \varphi \\ z = h \end{array} \right. \rightarrow \iint_{S_{\text{down}}} + \iint_{S_{\text{up}}} = 2 \iint_S (x+y+z) dx dy dz =$$

$$= 2 \int_0^{2\pi} d\varphi \int_0^H dh \int_0^h [r(\cos \varphi + \sin \varphi) + h] r dr = \frac{\pi H^4}{2}$$

$$S_{\text{up}}: z = H, x^2 + y^2 = z^2 \rightarrow \iint_{S_{\text{up}}} = \int_0^{2\pi} d\varphi \int_0^H H^2 r dr = \pi H^4$$

$$\iint_{S_{\text{down}}} = -\frac{\pi H^4}{2}$$

$$\sqrt{62} \quad \vec{n} = \frac{1}{\sqrt{3}} (1, 1, 1)^T, \quad \vec{m} = (0, 1, 0) \quad \cos(\vec{n}, \vec{m}) = \frac{1}{\sqrt{3}} \Rightarrow$$

$$\text{rot } \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & z^2 & x^2 \end{vmatrix} = -2z\vec{i} - 2x\vec{j} - 2y\vec{k}$$

$$\int_L = \iint_S (\text{rot } \vec{a}, \vec{n}) dS = -\frac{2}{\sqrt{3}} \iint_S (x+y+z) dS = -a^3$$

$$\sqrt{63} \quad L = \partial G, \quad G_0 = \text{Kugel} \cdot \text{oben} \cdot \text{cong.} \cdot \tau(0,0) \quad G_1 = G \setminus G_0$$

$$\iint_{\partial G_0 \downarrow} + \iint_{L \uparrow} = \iint_{\partial G_1 \uparrow} = \iint_S (\text{rot } \vec{a}, d\vec{S})$$

$$\vec{a} = \left(-\frac{y}{x^2+y^2}, \frac{x}{x^2+y^2}, z \right)^T, \quad \text{rot } \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} & z \end{vmatrix}$$

$$= (0, 0, 0)^T$$

$$\iint_{G_1} = 0 \rightarrow \iint_{L \uparrow} = \iint_{\partial G_0 \uparrow}$$

$$\begin{aligned} x &= R \cos \varphi \\ y &= R \sin \varphi \\ z &= -R(\cos \varphi + \sin \varphi) \\ dz &= -R(\cos \varphi - \sin \varphi) \end{aligned}$$

$$\int_0^{2\pi} (1 + R^2 \cos 2\varphi) d\varphi = 2\pi$$

$$\sqrt{73} \quad \vec{a} = (2x+3y, x+y+z, x+2y+3z)^T$$

$$\text{div } \vec{a} = 2+1+3=6$$

$$x = 2r \cos \varphi \cos \psi$$

$$y = 5r \sin \varphi \cos \psi \rightarrow \iint_S = 6 \iiint_G dx dy dz = 60 \int_0^{2\pi} d\varphi \int_0^{\pi/2} \cos \psi d\psi$$

$$z = r \sin \psi$$

$$\int_0^1 r^2 dr = 40\pi$$

$$S = S_{\text{oben}} + S_{\text{unten}}$$

$$x=0: \iint_{\Sigma} 3y dy dz = \int_0^{2\pi} \int_0^{2\pi} \left\{ \begin{array}{l} y=5r \sin \varphi \\ z=r \cos \varphi \\ r=5r \end{array} \right\} = \int_0^{2\pi} \int_0^{2\pi} 5r^2 \sin \varphi 5r dr d\varphi$$

= 0

$$\sum_{\partial K} = 40\pi - 0 = \underline{40\pi}$$

✓ 68 $\left\{ \begin{array}{l} \text{на } S: r=r_0, \operatorname{div} \vec{r} = 3 \end{array} \right.$

$$\iint_S (\vec{a}, \vec{n}) dS = \iint_S (\vec{a}, d\vec{S}) = \frac{1}{r_0^3} \iiint_G \operatorname{div} \vec{r} dx dy dz =$$

$$= \frac{1}{r_0^3} \cdot 3V_G = \underline{4\pi}$$

✓ 94 $\left\{ \begin{array}{l} \vec{a} = (y, -x, z)^T, \operatorname{rot} \vec{a} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & -x & z \end{vmatrix} = \\ = (0, 0, -2)^T \end{array} \right.$

$$\int_r (\vec{a}, d\vec{r})^* = \iint_S (\operatorname{rot} \vec{a}, d\vec{S}) = -2 \int_S dS = \underline{-4\pi}$$

✓ 104 $\left\{ \begin{array}{l} \operatorname{rot} \vec{a} = \vec{0} \end{array} \right.$

1) $G = \{x > 0\}$ - поверхн. огн. $\rightarrow$ га

2) $G = \{x \neq 0, y \neq 0\}$

$$\operatorname{rot} \vec{a} = \vec{\nabla} \times \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$\int_r (\vec{a}, d\vec{r}) = \frac{2I}{R^2} \iint_S (\vec{r}_0 + \vec{a}', d\vec{S}) = 4\pi I \neq 0 \rightarrow \underline{\text{нет}}$$

✓ 112

$G = R \setminus \{0\}$ - не явл. объемно огн. $\rightarrow$

• $\operatorname{rot} \vec{a} = \vec{0}$, G - поверхн. огн. $\rightarrow$ потенц.

• $\operatorname{div} \vec{a} = \frac{3}{r^3} - \frac{3}{r^4} \cdot r^3 = 0$

a) $z > 0$ - объем. огн. $\rightarrow$ конечна

b) $r > 0$ - не явл. обк. огн.

$$\iint_S \left(\frac{\vec{r}}{r^3}, d\vec{S} \right) = \frac{1}{R^3} \iint_S (\vec{r}, d\vec{S}) = \frac{1}{R^3} \iiint_G \operatorname{div} \vec{r} dx dy dz = \frac{4\pi}{R^3} \neq 0$$

- не конечна